

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech

Branch : Mechanical Engineering /Mechanical

Semester : III

Engineering (Sandwich) /Mechanical and Automation

Subject Code & Name: BTMC303 Thermodynamics

Max Marks: 60

Date: 10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
	A system in which energy can cross the boundary of the system but the mass cannot is known as	1/CO- 1	1
	a. open system b. Closed system c. Isolated System d. None of These		
	Which of the following is an intensive property of a thermodynamic system?	2/CO- 1	1
	a. mass 1/CO- 1 c. enthalpy d. density		
	When two bodies are in thermal equilibrium with a third body, they are also in thermal equilibrium with each other. This statement is called _____ of thermodynamics	2/CO- 1	1
	a. Zeroth Law b. First law c. second law d. third law		
	First law of thermodynamics deals with _____	2/CO- 2	1
	a. Conservation of mass 1/CO- 1 c. Conservation of energy d. Conservation of pressure		
	An increase in enthalpy leads to an increase in _____	1/CO- 2	1
	a. Increase in pressure b. Increase in internal energy c. Increase in volume d. Increase in mass		
	SI unit of enthalpy is _____	2/CO- 3	1
	a. Joule/kg 1/CO- 1 c. Joule/K d. K/kg		
	In Carnot cycle the heat is received at _____ Temperature.	2/CO- 3	1
	a. Lower 1/CO- 1 c. constant d. None of the above		

8	Gases of the same volume contain the same number of molecules. This is the description of				2/CO- 4	1
	a. Boyle's law	b. Dalton's law	c. Charles' law	d. Avogadro's law		
9	The throttling process is _____ process.				2/CO- 3	1
	a. reversible	b. irreversible	c. reversible or irreversible	d. None of these		
10	Which of the following is not true about ideal gas molecules?				3/CO- 4	1
	a. They have negligible size	b. They do not have attractive forces	c. They do not apply pressure	d. They move in random motion		
11	The value of pressure of water at a critical point is in bar...				4/CO- 5	1
	a. 223.5 bar	b. 212.25 bar	c. 235.5 bar	d. 221.2 bar		
12	With the decrease in pressure the latent heat of steam is....				2/CO- 5	1
	a. Decreases	b. increases	c. Constant	d. Increases or decreases		
Q. 2	Solve the following.					12
A)	Explain Intensive and Extensive properties with suitable examples.				2/CO-1	6
B)	Explain 'Thermodynamic equilibrium' of a system and state its significance.				2/CO-1	6
Q.3	Solve the following.					12
A)	Explain the first law of thermodynamics for (i) closed system undergoing a cycle (ii) closed system a change of state.				2/CO-2	6
B)	A gas undergoes a thermodynamic cycle consisting of the following processes: (i) Process 1-2: Constant pressure $p = 1.4 \text{ bar}$, $V_1 = 0.028 \text{ m}^3$, $W_{1-2} = 10.5 \text{ KJ}$. (ii) Process 2-3: Compression with $pV = \text{constant}$, $U_3 = U_2$ (iii) Process 3-1: Constant volume $Q_3 - U_3 = -26.4 \text{ KJ}$. There are no significant changes in KE and PE. (a) Sketch the cycle on p-V diagram. (b) Calculate the net work for the cycle in KJ. (c) Calculate the net heat transfer for the process 1-2 (d) Show that $\sum Q_{\text{cycle}} = \sum W_{\text{cycle}}$				3/CO-2	6
Q. 4	Solve Any Two of the following.					12
A)	Define the term COP as applied to a refrigerator and heat pump. Show: $\text{COP}_{\text{HP}} = \text{COP}_{\text{REF}} + 1$				4/CO-3	6

B)	A cyclic heat engine operates between a source temperature of 1000°C and sink temp of 40°C . Find the least rate of heat rejection per KW net output of the engine?	4/CO-3	6
C)	Show that Entropy changes in reversible process is Zero	4/CO-3	6
Q.5	Solve Any Two of the following.		12
A)	State 'Avagadro's Hypothesis' and from it determine the value of Universal gas constant.	1/CO-4	6
B)	Define specific heats. Derive relationship between C_p and C_v .	2/CO-4	6
	Derive expression for work done during polytropic process for a non-flow system	4/CO-4	6
Q. 6	Solve Any Two of the following.		12
A)	What is pure substance? Give suitable example. What is saturation pressure and saturation temperature? Explain it on T-S diagram.	2/CO-5	6
B)	A vessel having a capacity of 0.04 m^3 contains a mixture of saturated water and saturated steam at a temperature of 250°C . The mass of steam present is 9 kg. Find the following i. The Pressure ii. The mass iii. The specific enthalpy iv. the specific volume v. the specific entropy vi. The specific internal volume	4/CO-5	6
	Draw a P-T (Pressure-Temperature) diagram of Pure substance (Water) and explain sublimation curve and Fusion curve	4/CO-5	6
*** End ***			